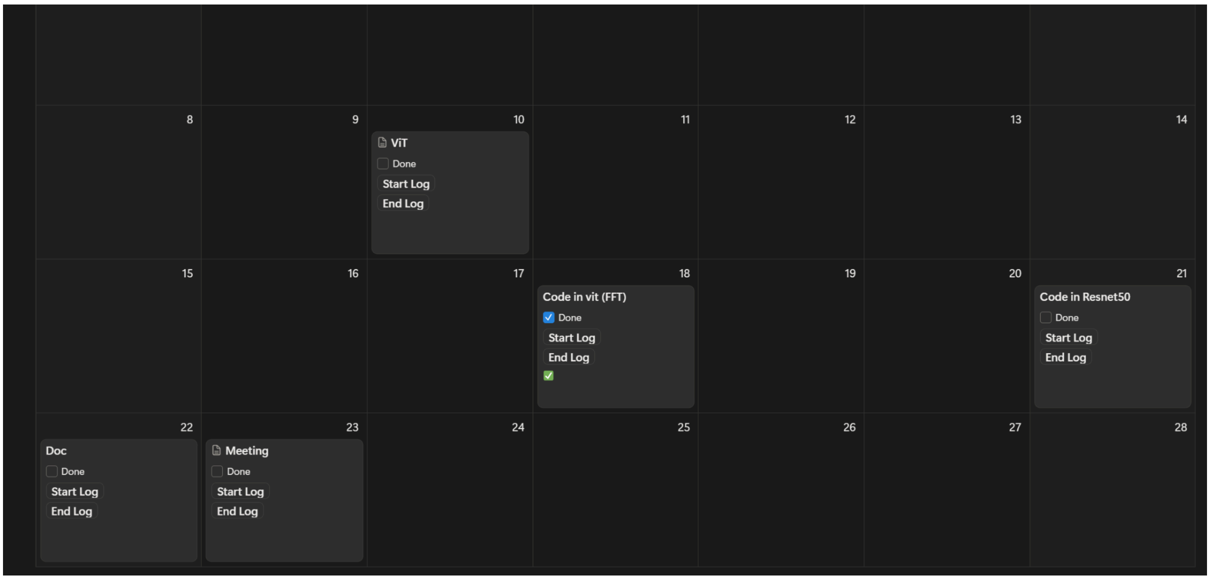
Reflection 1: Supervisor/project consultation

https://www.notion.so/Meeting-31096cbb797780aeab85de2b7d47f900?source=copy_link



First of all, in one of my meetings with my adviser (Ivy) about the detection project, we reviewed the direction and technical depth of my project on deepfake image detection. The primary focus was on determining which dataset to use to construct a reliable model to detect deepfake images. My adviser asked me to explain how my chosen dataset would be appropriate for the project since it would influence the accuracy of the model. When I was preparing my response, I realised that the selection of a dataset is not just a step in preparing for the technical aspect of the project, but is also a significant driver in determining the overall validity and reliability of the project.

Then, I was only concerned with whether my dataset was large and had both real images and AI-generated images. After the consultation, I can see that I also need to think about the diversity of the dataset, where the images were from, how they were generated, possible bias, and whether the dataset represents real-world deepfake detection situations.

Also, if the limitations of the dataset are too huge the model will only learn patterns unique to the dataset, instead of learning general deepfake features. The model will therefore perform great during testing, but will not generalize well when applied to unseen images. This is why I will describe the datasets more clearly in my report by providing the dataset source, how the dataset is structured, classes of reality/fake, preprocessing steps that occurred, and limitations.

Attacker Concern

During the consultation, Ivy raised another concern that if an attacker observed the features used by my model and attempted to modify fake images to avoid detection. This question raised my view on this project. I had been thinking of the model mainly as a classifier producing binary true/false results, but then recognised that the attackers are likely to be able to modify their behaviour based on how the detection system works in a real-world security environment. This could occur because the detection system is based on visible artefacts or frequency-based patterns of the image. Thus, an attack may use a post-processing technique to conceal those artefacts.

After the consultation, it changed how I approach the design of the project because I began to critically evaluate the overall design of my project. I was not just concerned with whether the model could tell whether an image was real or fake, but began asking if the model would be able to work if the inputs changed, whether the model could generalise to new examples of data, and if an attacker could bypass the model using different methods. This allowed me to link the technical aspects of my project with a relevant threat model from a security perspective. The consultation also influenced how I structured my report, in particular the dataset section, evaluation section and future directions section. Overall, the supervisor meeting has resulted in me moving from simply completing an image classification project to completing a project that takes into account the implications of developing a deepfake detection system that includes security awareness.